

A Reinforcement Learning Based Hyper-Parameter Generation System Guided by LLM-Powered Virtual Users

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Abstract

In e-commerce search systems, the personalized optimization of hyper-parameters associated with search scenario characteristics can significantly improve the quality of exposed item sequences. Recently, Reinforcement Learning (RL) algorithms have been applied to process complex scenario features and generate personalized parameters. Nevertheless, the high cost and risk of online exploration still present significant challenges for the training and deployment of RL agents. To address these challenges, a DDPG-based hyper-parameter generation system incorporating personalized search scenario features is proposed, and a virtual user behavior simulation system is developed based on Large Language Models (LLMs) for the exploration interactions of the RL agent. In addition, for improving the simulation performance of virtual users, a Chain of Thought for e-commerce scenario (e-CoT) is derived based on the counterfactual reasoning of real e-commerce user behavior. And the virtual user is fine-tuned based on a realistic dataset extracted from Alibaba e-commerce platform interactions, which also provides a critical foundation for developing and validating e-commerce search algorithms. Finally, both offline and online experiments demonstrate that the proposed hyper-parameter generation method can significantly enhance the performance of e-commerce search systems. These results are expected to provide valuable insights for deploying large-scale exploration algorithms in e-commerce search system.

CCS Concepts

• **Information systems** → *Learning to rank*.

Keywords

Hyper-parameter optimization, reinforcement learning, large language model, fine-tuning

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1 Introduction

In e-commerce search systems, the exposure item sequence is typically generated by the aggregation of multidimensional scores in the Learning to Rank (LTR) module. In order to match user preferences, it is essential to optimize the hyper-parameters based on user and search scenario features.

In recent years, Reinforcement Learning (RL) has been applied to personalized parameter optimization in recommender systems [1, 8, 13, 19]. For instance, Xi *et al.* combined transformer models with reinforcement learning to process long sequence information, effectively enhancing the long-term value of recommender systems [15]. However, user behavior in e-commerce search scenarios is influenced by multiple factors, including search queries, personal preferences, and potential intentions. Additionally, the risks associated with exploration actions in search systems are more significant than in recommender systems, and policy evaluation through online A/B experiments is highly costly. Therefore, developing efficient, reasonable, and safe methods for evaluating agent exploration remains an urgent challenge.

To support RL agent exploration, developing virtual behavior simulation environments represents a valuable research direction [12, 21]. Previous work by Shi *et al.* established a user behavior simulation system (Virtual-Taobao) using generative adversarial networks[12]. However, such approaches exhibit significant dependence on training data and struggle to capture complex user decision-making psychology due to reasoning limitations. In recent years, the emergence of Large Language Models (LLMs)[7, 9, 17] has introduced anthropomorphic reasoning capabilities that show promise for user behavior simulation. Recent studies have applied LLMs to estimate user preferences toward items[5, 10, 11, 18, 20]. Yet current LLM-based simulation methods typically lack structured decision-making processes, particularly the chain of thought (CoT) framework[14] needed for e-commerce scenarios. This highlights the need to develop a realistic CoT framework based on actual e-commerce user thinking diagram. Such advancement would significantly enhance LLM behavior simulation validity, thereby providing a more reliable foundation for RL agent training and ranking strategy evaluation.

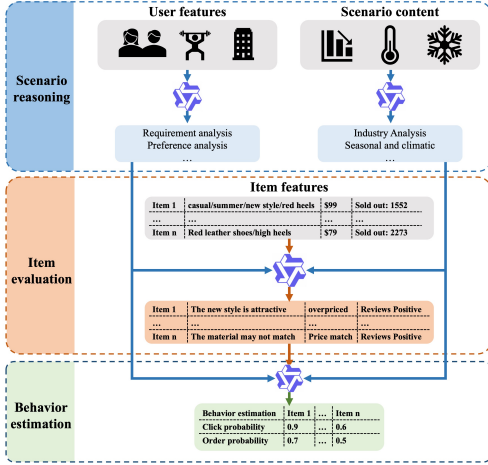


Figure 1: The thought process of e-CoT in LLM-powered virtual user behavior simulation system

To achieve personalized hyper-parameter optimization in search scenarios, we propose a comprehensive RL-based framework for e-commerce search systems. The primary contributions include:

- In order to assess the ranking quality reasonably, a virtual user simulation system is developed, where a LLM acts as customer and interacts with the items virtually.
- In order to improve the performance of virtual user, an e-commerce decision-making CoT (e-CoT) framework is derived from real user behavior diagram. And the LLM is fine-tuned based on Real platform user log dataset.
- A personalized hyper-parameter optimization agent is developed using the Deep Deterministic Policy Gradient (DDPG) algorithm. And a comprehensive multi-objective reward function designed to enable efficient and effective training of the DDPG agent.

2 Problem and Formulation

In e-commerce search systems, candidate items are typically evaluated across multiple dimensions, including relevance, conversion efficiency, and other performance metrics. These diverse feature scores are aggregated into a final ranking score through a learning-to-rank (LTR) module, formulated as:

$$S = \frac{\sum_{i=1}^n \omega_i s_i}{\sum_{j=1}^n \omega_j}, \quad (1)$$

where the importance of each score is controlled by corresponding weight parameters ω_i , which affects the characteristics of the ranking results. Consequently, personalized parameter optimization can significantly enhance both the quality of the exposure sequence and the overall performance of the search scenario.

3 Methodology

3.1 LLM-Powered Virtual User Behavior Simulation System

To safely and efficiently evaluate the exploration behavior of the DDPG agent during training, a virtual user behavior simulation system is implemented. In this system, the LLM acts as an e-commerce

buyer and replicates the characteristics of sampled users, including static attributes (e.g., gender) and behavioral patterns (e.g., browsing history). The input context combines search queries, scenario meta-data, and promotional campaigns. Candidate items are presented to the LLM as an ordered list mimicking real-world e-commerce platforms, complete with product titles, pricing, and user reviews. By analyzing user profiles, contextual signals, and item features, the virtual behavior simulation system sequentially predicts click-through and purchase probabilities for each item as follows:

$$P_{click} = [p_1^c, p_2^c, \dots, p_n^c], P_{order} = [p_1^o, p_2^o, \dots, p_n^o] \quad (2)$$

Given the complexity of e-commerce search and interaction behaviors, the counterfactual reasoning of e-commerce behavior logic is carried out based on Qwen3-235B. With a batch of real user behavior logs, the complete e-commerce decision-making CoT (e-CoT) is summarized to enhance the logical coherence and consistency of simulated virtual behavior. As illustrated in Figure 1, the e-CoT comprises three core reasoning nodes: scenario pre-reasoning, item evaluation, and behavior estimation. In the phase of scenario pre-reasoning, user requirements and preferences are estimated by analyzing static characteristics and historical behavioral sequences. Additionally, by incorporating the search query, the expected product characteristics are inferred as intermediate information for the following inference stage. In parallel, the factors that potentially influence user decisions, such as discount information, seasonal factors, and other contextual variables, are comprehensively analyzed to support the following decision-making process. On the basis of scenario pre-reasoning, candidate items in the exposure sequence are evaluated from multiple aspects such as relevance, matching degree of user preference, and so on. Finally, the unified virtual behavior reasoning is realized by integrating the results of scenario pre-reasoning and the evaluation reasoning candidate items.

For further improving the inference performance of LLM-based virtual users, a dataset including 10k search scenario, 100k item information, and 10k user information is constructed from the Alibaba e-commerce platform. On this basis, the LLM is fine-tuned based on Group Relative Policy Optimization (GRPO) algorithm, where the virtual behavior generation reward is calculated based on the cross-entropy function as follows:

$$L = \frac{-\sum_N (y_i * \log(p_i) + (1 - y_i) * \log(1 - p_i))}{N} \quad (3)$$

3.2 RL-Based Hyper-Parameter Optimization

In order to realize the personalized hyper-parameter configuration according to complex and high-dimensional user characteristics in e-commerce search system, a complete continuous parameter optimization framework is developed based on the Deep Deterministic Policy Gradients algorithm (DDPG)[2, 6], as illustrated in Figure 2, with detailed descriptions provided in the subsequent sections.

The input state of the agent is defined by a N -dimensional vector representing encoded user features. Specifically, continuous attributes such as user age are directly input as raw values, binary attributes such as gender are encoded as 0 or 1, and discrete attributes such as user profile classifications are mapped to corresponding integer codes. Besides, the search query is encoded as a predicted category id and embedded into the input state. The

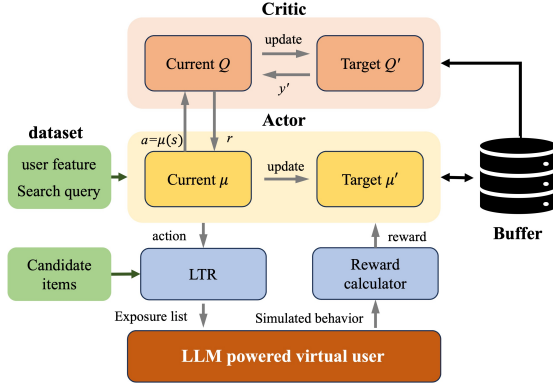


Figure 2: The construction of RL-based hyper-parameter optimization system guided by LLM-power virtual users.

action space is the fusion weight with continuous values in LTR module. And the range action is limited to between 0.001 and 20.0 depending on the magnitude of scores to be integrated.

In particularly, a comprehensive and efficient reward function is designed as following. Assume $I_1, \dots, I_n$ is an ordered list of exposure items, with the probabilities of click $P_{click} = [p_1^c, p_2^c, \dots, p_n^c]$ and order $P_{order} = [p_1^o, p_2^o, \dots, p_n^o]$, respectively. The reward of a search trace is calculated as follows:

$$r = \frac{\sum_{i=1}^n d^{k_i} \cdot price_i \cdot (\alpha p_i^c + \beta p_i^o)}{\sum_{j=1}^n price_j} \quad (4)$$

where the price of item is normalized. k_i indicates item i is positioned at the k -th rank. d^{k_i} is the discount coefficient for the exposure position effect, and d ranges as 0.1.

The designed reward function implied the contribution of clicks, conversions and revenue, so as to avoid complicated statistical calculations and improve the evaluation efficiency in the training process. In addition, the reward contribution of each item is derived from a continuous probability value rather than a boolean value, which improves the integrity of the evaluation.

Additionally, to improve reward consistency, the ranking results generated by the exploration actions are evaluated by the simulation system (r_π) along with the original presentation sequence (r_o). And the non-linear relative reward as (5) is applied as the final training reward.

$$R = \begin{cases} -e^{(r_o - r_\pi)} & \text{if } r_\pi < r_o \\ r_\pi - r_o & \text{if } r_\pi > r_o \end{cases} \quad (5)$$

4 Experiments and Analysis

4.1 The Evaluation of User Behavior Simulation System

The simulation system under evaluation is implemented based on the Qwen3-8B model[16]. The evaluation data are sampled from the constructed dataset, where the actual user interaction behaviors serve as the baseline. For quantitative analysis, the HitRate in top- k ($HitRate@k$) is calculated to evaluate the estimated probability of interaction, and the average differences in click price ΔC_{price} and position $\Delta C_{position}$ are acquired to evaluate the behavioral similarity between virtual and real user.

Table 1: The evaluation of virtual user simulation system

Method	ΔC_{price}	ΔC_{pos}	$HitRate@5$	$HitRate@10$
virtual user	3.4%	1.9%	69.1%	78.8%
LLM w/ GRPO	8.6%	7.5%	53.7%	77.5%
LLM w/ e-CoT	6.2%	4.4%	58.5%	78.0%
LLM	10.1%	8.8%	50.9%	69.1%

As shown in Table 1, the inference performance of the virtual user model is significantly improved after combining the e-CoT and GRPO fine-tuning, obtaining higher $HitRate@k$ and lower simulation error in ΔC_{price} and ΔC_{pos} . Furthermore, to quantify the consistency of the LLM-generated behavior probabilities, the Pearson Correlation Coefficients (PCC) [3] of the original LLM model and virtual user model are calculated to assess test-retest reliability [4]. The results show that e-CoT guidance and GRPO fine-tuning can substantially improve output consistency (PCC: 0.71 $\rightarrow$ 0.86), which is crucial to improve the iteration stability of RL training.

4.2 The Evaluation of DDPG-based Hyper-Parameter Optimization System

In this section, the DDPG-based hyper-parameter agent is trained with behavior labels from the e-commerce dataset and virtual behavior generated by the LLM-based simulation system, respectively. The training process consists of 4×10^5 episodes, with a learning rate of 1×10^{-4} and a batch size of 32. For comparative analysis, the Genetic Algorithm (GA) is implemented for comparison. The Area Under Curve (AUC) and $HitRate@k$ are calculated as quantitative metrics, with a rule-based parameter optimization method serving as the baseline.

Table 2: The evaluation of RL-based optimization system

Method	AUC	$HitRate@5$	$HitRate@10$
Baseline	0.689	54.3%	78.3%
GA & dataset	0.717	59.8%	87.3%
RL & dataset	0.733	60.1%	89.2%
RL & LLM	0.739	60.2%	89.9%
RL & LLM & e-COT	0.758	62.8%	90.3%

As shown in Table 2, both the GA and the proposed DDPG-based optimization method yield performance improvements. However, the GA exhibits relatively weaker enhancement due to its limited personalized optimization capability compared to the DDPG approach. The analysis further reveals that interaction behavior quality significantly impacts model performance. DDPG agents trained with LLM-generated virtual behaviors achieve more substantial improvements in both AUC and $HitRate@k$ metrics, attributable to the superior stability and rationality of simulated behaviors compared to original dataset labels. Furthermore, the application of e-CoT further improves the consistency of virtual behavior, enabling the DDPG agent to achieve performance gains of 10.0% in AUC and 15.3% in average $HitRate$, respectively.

In addition, the fixed behavior labels in the dataset fail to capture user decision deviations caused by changes in item exposure positions within reordered sequences. In contrast, the LLM-based

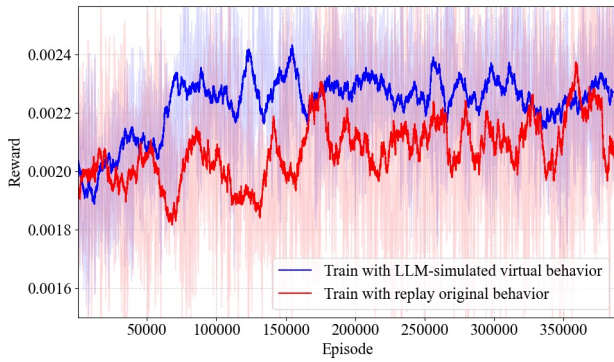


Figure 3: Training loss with different simulation environment

behavior simulation system generates more rational and stable interaction behaviors that demonstrate stronger correlations with user characteristics. Consequently, the convergence speed and training stability of the agent are significantly improved, as demonstrated in Figure 3.

4.3 Online A/B Experiments

To further evaluate the online effectiveness of the hyper-parameter optimization system, an A/B experiment is conducted on the Alibaba e-commerce search engine. After a continuous one-month trial, the conversion rate (L2O) and Gross Merchandise Value (GMV) increased by 0.84% and 3.06%, respectively. And it is more significant than the improvement obtained by the evolutionary algorithm, with 0.11% increase in L2O and 1.15% increase in GMV. These results demonstrate that the LLM-powered user behavior simulation environment exhibits strong consistency with real-world conditions, and the DDPG-based personalized optimization algorithm effectively aligns with the habits of online users.

5 Conclusion

In this paper, a personalized hyper-parameter optimization algorithm considering search scenario characteristics is proposed based on DDPG To improve the performance of e-commerce search system, and an efficient reward function is designed to improve user experience and platform revenue. In addition, a LLM-based virtual user behavior simulation system is developed to evaluate the exploration behavior of DDPG agent while training. Furthermore, for improving the rationality and consistency of LLM reasoning, a e-commerce decision-making e-CoT is summarized with , and GRPO fine-tuning is applied based on real Alibaba e-commerce dataset. Finally, both the offline and online experiments validate the effectiveness of the proposed optimization algorithm and simulation environment.

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